Three further reductions for the conditional LOCAL42 route

Research and independent audit in the user's fork

27 October 2026

Abstract

We give an elementary replacement for all 65 zero-loss support calls, an exact three-candidate formula for the LOW curvature supremum, and a tighter resource argument that removes two selected LOW producers. These are conditional components, not a proof of the sharp iid Berry–Esseen conjecture. The full acquisition, equality-compatible attainment, contact, local-comparison and finite-base obligations remain. No original certificate, target or production checker is changed. A separate one-pass execution prototype is discussed only at its tested protocol scope.

1 Scope and preserved premises

The user's fork is `kolodziejek/berry-esseen-certificates`. The immutable reference and this continuation's starting commit are, respectively:

```
6346fc1b4432bc5982822aa696eb566ed9e1e1ee
42433b448d9389359e642a3384a949ffdd87b903
```

Exact source paths, hashes and literal line locations are in the companion notes:

```
elementary-support.md
local-kernel.md
tighter-resources.md
```

Their small checkers read the pinned source archive without executing its native arithmetic kernels.

The earlier reduction retains a genuine sharp base through 41. At a least index $n \geq 42$ with $C_n \geq C_{\mathrm{E}}$, both required predecessors are sharp, the global predecessor decrement is zero, and support-free smoothing gives

$$n(C_n - C_{\mathrm{E}}) < \frac{3003}{10250} < 0.3, \qquad C_n < \frac{73}{175} < 0.4226.$$

Both deleted-score caps are below 0.4253. Those scalar conclusions alone do not establish LOCAL42. In particular they do not imply zero conditional noise, compact support, attainment, or a numerical certificate.

The support result below applies to an attained positive closed maximizing pair (P, H) with standardized iid marginal, third absolute moment $\rho = E|X|^3$, and signed moment $\beta = E[X|X|]$. All moment and same-law contact premises are retained. The resource result is applied only *after* the corresponding confinement and acquisition. Its extension from the literal finite 42–48 source scope requires the separate uniform42 bridge; the floor 42 alone does not prove that extension.

2 Elementary support without a positivity mesh

Throughout this section, the actual integer satisfies $n \geq 42$ and the attained score satisfies $C_n \geq C_{\mathrm{E}}$. Set $v = n - 1$, $s = \sqrt{n/v}$ and $F_v(x) = \Phi(x/\sqrt{v})$. The retained contact inequality on the complete closed support is

$$0 \leq H_n(H,t) + C_n\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\}, \tag{1}$$

where

$$e_n = n\big(\sqrt{n/(n-1)} - 1\big) - \tfrac{1}{2} < \frac{0.384}{n},$$

and the exact Gaussian expression is

$$\frac{H_n(H,t)}{n^{3/2}} = F_v(H-t) - F_{v+1}(H) + tF'_{v+1}(H) - \frac{t^2-1}{2}F''_{v+1}(H).$$

The same-owner predecessor score is at most the global predecessor score; at zero global decrement the discarded signed-loss term is nonpositive.

2.1 Fourth-order expansion with uniform remainder

Using $\partial_v F_v = F''_v/2$, expand $F_v(H-t)$ spatially through order four, F_{v+1} through second order in variance, and its first and second spatial derivatives through first order in variance. The order-zero, linear and quadratic spatial terms cancel. The remaining cubic and quartic coefficients are $(-t^3+3t)/6$ and $(t^4-6t^2+3)/24$. Thus, for $t = \pm u$,

$$\frac{H_n(H,\pm u)}{s^3} = \pm A(1-z^2)\phi(z) + B\phi'''(z) + R, \qquad z = H/\sqrt{v}, \tag{2}$$

where

$$A = \frac{u^3-3u}{6}, \qquad B = \frac{u^4-6u^2+3}{24\sqrt{v}}.$$

For $u \geq 5/2$, both are positive. The four Taylor remainders, after division by s^3, have bounds

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120v}, \quad \frac{\|\phi^{(5)}\|_\infty}{48v^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v}, \quad \frac{(u^2-1)\|\phi^{(5)}\|_\infty}{16v^{3/2}}.$$

Gaussian Fourier inversion gives $\|\phi^{(4)}\|_\infty \leq 3\phi(0) < 6/5$ and $\|\phi^{(5)}\|_\infty \leq 8/\pi < 51/20$. Therefore

$$|R| \leq E_v(u) := \frac{u^5/100 + 3u/20}{v} + \frac{51(3u^2-2)}{960v^{3/2}}. \tag{3}$$

Every bound is uniform in the raw threshold H. No seventh derivative of F_v, sixth derivative of ϕ, or signed fifth-order tangent is used. The rational lower bound $\pi > 281476/89625 > 160/51$ suffices for these constants and $\phi(0) < 2/5$.

Lemma 1 (Two all-threshold envelopes)**.** *For $u \geq 5/2$,*

$$H_n(H,\pm u) \leq s^3\{\phi(0)(A+2B) + E_v(u)\}.$$

If also $B \leq A/4$, then

$$H_n(H,\pm u) \leq s^3\left\{\phi(0)\left(A + \frac{9B^2}{4A}\right) + E_v(u)\right\}.$$

Both assertions hold for every real H and both signs separately.

Proof. The first follows from $\|(1-z^2)\phi(z)\|_\infty = \phi(0)$ and $\|\phi'''\|_\infty < 2\phi(0)$. For the positive sign and $|z| \le 1$, the retained terms in (2) are at most $\phi(0)(A - Az^2 + 3B|z|)$; completing the square proves the sharper bound. For $|z| > 1$ the first term is nonpositive and the second is at most $2B\phi(0) \le A\phi(0)/2$. For the negative sign, its first term is at most $2e^{-3/2}A\phi(0) < A\phi(0)/2$, and the second is at most $2B\phi(0) \le A\phi(0)/2$. Add the uniform remainder. □

2.2 Rational boundary inequalities and full rays

Use

$$S = 81/80, \quad \ell = 6, \quad c_0 = 819/2000 < C_{\mathrm{E}}, \quad H_4 = u^4 - 6u^2 + 3,$$

so $s \le S$, $\sqrt{v} > \ell$ and $e_n < 8/875$. Define

$$\begin{aligned}
E(u) &= \frac{u^5}{4100} + \frac{3u}{820} + \frac{17(3u^2-2)}{78720},\\
Q_{r,b}(u) &= -u^3 + \tfrac{3}{2}ru^2 + 3bu + \tfrac{8}{875}r,\\
P_{r,b}(u) &= \tfrac{2}{5}S^3(A + H_4/72) + S^3E(u) + c_0Q_{r,b}(u),\\
P^\sharp_{r,b}(u) &= \tfrac{2}{5}S^3\left(A^2 + \frac{H_4^2}{256\cdot 41}\right) + S^3E(u)A + c_0Q_{r,b}(u)A.
\end{aligned}$$

For acquired $\rho \le r$, $|\beta| \le b$, the contact coefficient is at most $Q_{r,b}$. Check $Q_{r,b} < 0$ before replacing its actual multiplier C_n by the lower constant c_0. Lemma 1 then makes $P < 0$ sufficient at a boundary; $P^\sharp < 0$ is sufficient when $B \le A/4$.

To extend a boundary to a whole ray, put

$$L_{r,b}(u) = \tfrac{2}{5}(u^2/2 + 1/2 + 8/875) + c_0Q'_{r,b}(u).$$

The retained direct derivative bound is $|\partial_t H_n| \le (2/5)(t^2/2 + 1/2 + e_n)$. If Q, Q', Q'', L, L', L'' are negative at U, their required signs persist for $u \ge U$, since $Q''' = -6$ and $L'' = 2/5 - 6c_0 < 0$. The contact upper bound for each sign then decreases strictly along the whole ray. This includes endpoints and atoms.

Initial case. Take $R = 96/25$, $r = 1841/1000$, $b = 321/500$. The retained moment inequalities give

$$\rho < \min\left\{\frac{.3031\cdot .646}{c_0 - .3031}, \frac{.3322\cdot .429}{c_0 - .3322}\right\} = \frac{7361}{4000} < r.$$

The sign-moment inequality gives $|\beta| < b$, using

$$(1+b^2)^2 - r^2(1-b^2) = \frac{93288849}{50000000000} > 0.$$

At $u = R < 6$, $H_4 = 6uA - 3(u^2-1) < 36A$ and $B < H_4/144 < A/4$. Exact arithmetic gives

$$-P^\sharp_{r,b}(R) = \frac{2214258283536204102723}{10250000000000000000000} > \frac{1}{5}.$$

All six ray signs are negative, proving support in $(-R, R)$.

Common LEFT bridge. Take $R = 121/40$, $r = 143/100$, $b = 93/200$, outwardly enclosing the acquired common moment maxima 71458741/50000000 and 46459359/100000000. Then

$$-P_{r,b}(R) = \frac{2533499510174559}{2149580800000000000} > \frac{1}{100}.$$

All six ray signs are negative.

A continuous family for the other 63 calls. All remaining acquired parameter tuples satisfy

$$\frac{5}{2} \le R \le \frac{111}{40}, \qquad \rho \le r_R = \frac{7}{20}R + \frac{1}{3}, \qquad |\beta| \le b_R = \frac{3}{8}R - \frac{3}{5}.$$

Let $p(R) = -P_{r_R,b_R}(R)$ and $w = R - 5/2$. The coefficients of $p(R) = \sum_{j=0}^{5} a_j w^j$ are

j	a_j
0	123579925563/1343488000000
1	−467155860219/1679360000000
2	705959271/13434880000
3	8703567/167936000
4	−7499223/839680000
5	−531441/2099200000

With $h = 11/40$, their signs imply throughout $0 \le w \le h$

$$p'(R) \le a_1 + 2a_2 h + 3a_3 h^2 = -\frac{797749885413}{3358720000000} < 0.$$

Consequently

$$p(R) \ge p(111/40) = \frac{4403736280322409}{2149580800000000000} > \frac{1}{50}.$$

For rays, freeze each pair (r_R, b_R). The common larger caps $r_* = 131/100$, $b_* = 441/1000$ have at $u = 5/2$ the six signs

$$-\frac{16991}{700000}, \quad -\frac{3801}{500}, \quad -\frac{1107}{100}, \quad -\frac{11615533}{7000000}, \quad -\frac{706633}{200000}, \quad -\frac{2057}{1000}.$$

They majorize the corresponding signs and derivative bounds for every fixed pair. Combining the negative boundary with these complete rays proves all 63 calls. Continuing the affine caps along an infinite ray would be invalid.

Theorem 1 (Conditional elementary support replacement)**.** *For an actual integer $n \ge 42$ and attained score $C_n \ge C_{\mathrm{E}}$, under the retained contact, zero-loss and ordered same-law acquisition hypotheses, all 65 support conclusions used by the zero-loss LOCAL42 route follow from the preceding three cases. The old radial Bernstein cells, the newer 14-group/672-coefficient support certificate, the signed fifth-order identity, its fourth-derivative lower tangent and its associated domination gates are unnecessary for this alternative support component.*

The input bindings and eighteen common ray signs are checked by the script

`check_elementary_support.py`

These arithmetic facts do not prove the acquired moment inequalities themselves.

3 Exact curvature maximization and safe transport

For real b, $1/2 < c \le 1$ and $[l, h] \subseteq [-1/2, 1/2]$, define

$$f(u) = \frac{(b+u-1)_+}{(c-u)^2} + \frac{(b+u)_+}{[c+\min(u, c/3)]^2}, \qquad u_* = c - 2b.$$

All denominators are strictly positive. In the kernels $c = 1 - d_L$ and $b = p_+ - \min(H/\ell)$, where the minimum uses all four signed endpoint quotients. The variable b is not a probability and cannot be freely reflected.

Theorem 2 (Exact three-candidate supremum)**.** *The supremum of f over $[l, h]$ is the maximum of $f(l)$, $f(h)$, and the value $1/[4(c-b)]$ if*

$$c > b, \qquad l \le u_* \le h, \qquad u_* \le c/3, \qquad u_* \le 1 - b.$$

An ineligible third value is omitted. The formula includes singleton intervals, coincident kinks and $c = 1$.

Proof. Write $f = r + g$ for the two summands. On $b + u > 1$,

$$r'' = \frac{2[3(b-1) + 2u + c]}{(c-u)^4} > 0,$$

since its bracket is $3(b+u-1) + (c-u) > 0$. Where $u < c/3$ and $b + u > 0$,

$$g' = \frac{c - 2b - u}{(c+u)^3}, \qquad g'' = \frac{2(3b + u - 2c)}{(c+u)^4}.$$

If also $r > 0$, then $3b + u - 2c > 3 - 2u - 2c \ge 0$, so f is convex there. For $u > c/3$, g is convex piecewise linear. At each of $-b$, $1-b$ and $c/3$, the derivative has a nonnegative jump; simultaneous jumps add. Thus none is a strict interior maximum.

Where $r = 0$, either $g = 0$, or below $c/3$ its derivative changes from positive to negative only at $u_* = c - 2b$. Its positive numerator condition is $c > b$, and $r = 0$ there is equivalent to $u_* \le 1 - b$. Its value is $1/[4(c-b)]$. Above $c/3$, g is nondecreasing. These cases locate a global maximum among the stated candidates. Continuity handles boundaries and collisions; constant segments contribute a value already attained at an endpoint or a listed maximizer. □

For an old grid interval $[a, d]$ and every $u \in [a, d]$,

$$f(u) \le \frac{(b+d-1)_+}{(c-d)^2} + \frac{(b+d)_+}{[c+\min(a, c/3)]^2}.$$

Therefore the exact maximum is at most the old 64-interval outward bound. For fixed b, u, f is nonincreasing in c. CENTRAL's first cached band has the smallest c, so its cached envelope also dominates later exact values; this requires the original band order and joint-gap relation. LEFT retains its band-dependent cache.

3.1 Why the final correction needs a zero clamp

The curvature improvement can make the recomputed inverse-index correction negative. One must not assume the old assertion $I > 0$ remains true. Within a fixed positive-noise cell, the actual forms are

$$B(\lambda) = z + \frac{q}{a\lambda + G^2} + s\lambda - t, \qquad I_{\mathrm{new}}(\lambda) = A_0 + \frac{A_1}{\lambda}, \quad q \geq 0,\ a > 0,\ A_1 \geq 0.$$

Only the curvature contribution to A_0 decreases. Put $I_+(\lambda) = \max(0, I_{\mathrm{new}}(\lambda))$. For $n \geq N$,

$$mB(\lambda) + \frac{I_{\mathrm{new}}(\lambda)}{n} \leq mB(\lambda) + \frac{I_+(\lambda)}{N}.$$

Both B and the right-hand expression are convex for $\lambda > 0$, including a possible interior zero-clamp kink, since $m > 0$. Endpoint checks thus cover the closed interval. At an accepted old endpoint,

$$I_{\mathrm{new}} \leq I_{\mathrm{old}},\ I_{\mathrm{old}} > 0 \quad \Longrightarrow \quad 0 \leq I_+ \leq I_{\mathrm{old}}.$$

The original strict target $mB + I_{\mathrm{old}}/N < -1/1000$ is preserved. The separate $B < 0$ gate remains necessary to use the mass lower bound.

This is conditional transport from an accepted old cell, not a full replay or a suppression of an old assertion. A new implementation replaces 64 candidate values by at most three per curvature cache miss. All marked-tail, normalization, mass, joint-resource and other comparison work remains.

4 Retaining the score-excess charge

Let the actual integer satisfy $n \geq N = 42$. Put $c = C_{\mathrm{E}}$, with actual score $C \geq c$, $\rho = E|X|^3 \geq 1$, and $\kappa = 997/15000$. Retain the source phase resource

$$n(C - c)\rho + \kappa n D + n^{3/2} L \leq \mathcal{B}(H), \qquad L \geq 0, \tag{4}$$

with its original same-law geometry, support, signed phase, concentration, moment and deleted-score hypotheses. In particular the conserved phase curvature is not charged twice.

For each of two acquired moment corners j and each phase sign, its quadratic upper envelope is

$$A_j(C) + F_{j,\pm}(C)|H| - Q_j(C)|H|^2.$$

Here A_j, Q_j are affine in C, and each signed F is a finite maximum of affine functions plus a score-independent slope. On the endpoint-score interval $[C_0, C_1]$, its chord $\widehat{F}$ majorizes F. The supremum of these chord quadratics is

$$G(C) = \max_j \left\{ A_j(C) + \frac{\max(0, \widehat{F}_{j,+}(C), \widehat{F}_{j,-}(C))^2}{4Q_j(C)} \right\}.$$

Positive endpoint Q_j gives positivity throughout. As a supremum of affine functions of C indexed by j, sign and $|H| \geq 0$, G is convex.

Lemma 2 (Two-interval score charge)**.** *Suppose $C_0 < c < \chi < C_1$ and the actual score is at most C_1. Then (4) implies*

$$\kappa n D \leq B_* = \max\{G(C_0), G(\chi), G(C_1) - N(C_1 - \chi)\}.$$

Proof. For actual $C \in [c, \chi]$, discard its nonnegative excess charge and bound $G(C)$ by the endpoints of the larger formal interval $[C_0, \chi]$. For $C \in [\chi, C_1]$, $n(C-c)\rho \geq N(C-\chi)$, using only $\rho \geq 1$. The convex function $G(C) - N(C-\chi)$ is bounded above by its endpoint values. Take the maximum. This includes $C = c$ without assigning a positive charge at equality. □

For LEFT, $C_0 = .4095$, $C_1 = .4226$ and $\chi = .40974$. Exact arithmetic on all 45 rows gives $B_* = G(\chi)$. For completeness, with $s_k(x) = \sum_{i=0}^{k-1} (-1)^i x^{2i+1}/(2i+1)$, the alternating-series bounds and Machin's tangent-addition identity give

$$16s_6(1/5) - 4s_1(1/239) < \pi < 16s_5(1/5) - 4s_2(1/239).$$

Together with $3.162277660 < \sqrt{10} < 3.162277661$ and $c^2 = (19 + 6\sqrt{10})/(72\pi)$ these prove $C_0 < c < \chi$ using rationals.

4.1 Keep one joint residual

The old weighted bound is $nD \leq N d_{\text{old}}$; define

$$d_{\text{new}} = \min\{d_{\text{old}}, B_*/(\kappa N)\}.$$

This is a bound on nD, not a bare bound on D subsequently rescaled. Retain the acquired geometry Θ, A, B and closed probability panel $[l, u]$. Write $s_l = 3.162277660$, $s_u = 3.162277661$, and set

$$\begin{aligned}
v_* &= \max_{p \in [l,u]} p(1-p), \\
\delta_* &= \max\{0, (4 - s_u)/2 - u,\ l - (4 - s_l)/2\}, \\
b_* &= (6 + 2s_l) \frac{\sqrt{(1-\Theta)^3}_{\text{lower}}}{\sqrt{v_*}_{\text{upper}}} \delta_*^2, \\
K &= \min\{3(s_l+1)B,\ 3(s_l+3)A\} > 0, \qquad r = d_{\text{new}} - b_*.
\end{aligned}$$

The retained same-law inequality is

$$nD \geq N b_* + 6N(1-\Theta)(A+B)d_{\text{in}} + \frac{K(1-\Theta)}{v_*}\lambda.$$

If $r < 0$, the row is impossible. If $r > 0$, put

$$\Lambda_{\text{new}} = \frac{N v_* r}{K(1-\Theta)}, \qquad d_{\text{new,in}} = \frac{r}{6(1-\Theta)(A+B)}.$$

Then the *single* residual gives

$$\frac{d_{\text{in}}}{d_{\text{new,in}}} + \frac{\lambda}{\Lambda_{\text{new}}} \leq 1.$$

It is not spent independently on noise and inward displacement. No tested row has $r = 0$; that case would require a separate zero-noise join.

CENTRAL exports upward-rounded A, F and downward-rounded Q, and only an outward raw score quantity. Its source rounding contract gives $\text{raw}_{\text{exact}} \in [\text{raw}_{\text{out}} - 2 \cdot 10^{-6}, \text{raw}_{\text{out}}]$. Accordingly its high score endpoint belongs to

$$[\,\min(.4247, \chi + (\text{raw}_{\text{out}} - 2 \cdot 10^{-6})/42), \min(.4247, \chi + \text{raw}_{\text{out}}/42)\,].$$

Enclose the chord weight over this interval; use upper A, F, lower Q, and the lower high-score endpoint in the subtraction in Lemma 2. This gives valid outward budgets for all 47 central rows.

4.2 Exact removable sectors

The following decimal ceilings are rounded upwards from exact fractions. Rows are zero-based. The full 92-row scan establishes all newly crossed thresholds among $.05, .08, .10, .12$.

Row	New conclusion	Consequence after acquisition
LEFT 0	$r < -.0003$	Entire final row excluded; TINY comparison unnecessary.
LEFT 5	$\lambda \leq .049944478$	MID domain above .05 empty.
LEFT 8	$\lambda \leq .077273055$	Both signed LOW producers above .08 unnecessary.
LEFT 10	$\lambda \leq .094055578$	Domain above .10 empty; two profile calls on $[.10, .12]$ unnecessary.
LEFT 13	$\lambda \leq .117287323$	Domain above .12 empty; two profile calls on $[.12, .30]$ unnecessary.
CENTRAL 28	$\lambda \leq .099809552$	Comparison sector above .10 empty; profile already inherited.

Restriction of the old complete cover to these smaller same-law domains preserves signs, nearest counts, closed phase seams and probability endpoints. Each removed closed noise band has its lower endpoint strictly above the new ceiling. The separate $\lambda = 0$ Bernoulli theorem remains. LEFT 6, 9 and 14 and CENTRAL 38 still exceed their relevant thresholds; this method does not justify deleting them.

Thus a newly named upper-module contract could use 150 LOW producers (78 CENTRAL, 72 LEFT), with 19 native obligations still retained. The candidate task count becomes 169, before new joins. This is not a count for the whole repository or a measured runtime saving. In particular, row 0 is excluded only after its final resources have been acquired: none of its acquisition or confinement ancestors is removed.

5 One-pass protocol and compatibility boundary

The old selected plan evaluates 152 LOW configurations in producers and evaluates them again in binders. The separately named prototype `onepass_low.py` retains the original 152 configurations, source pins, serialized rows, index hashes, coverage and nine binder controls per family. It owns fresh evaluator outputs in memory, runs the original non-replay binder validations, and compares read-back bytes and summaries to those fresh outputs. It does not accept user-supplied rows or receipts as a replacement for fresh evaluation.

Synthetic fixtures using the small original binder bodies produce identical legacy rows, indices and summaries with 152 rather than 304 evaluator calls; 14 further rejection controls pass. The native source-loading and full arithmetic path has not been executed, so this is protocol evidence, not a production acceptance result.

The curvature formula and the tighter resources change numerical outputs or the selected labels. They therefore do *not* satisfy the prototype's old output/index contract. Combining the three changes requires a new, complete contract and independent review. Their potential savings must not be multiplied into an unsupported overall speedup.

6 Evidence and remaining obligations

The component reports and `runs/reduction2-20261027/` separate mathematical and execution status. The support checker supplies exact three-case inequalities, all 65 source bindings and negative controls. The curvature checker tests 3,648 calls on the actual manifest grids, with 62,942 exact point diagnostics; these finite checks supplement the continuous proof, not replace it. Its first workload count used defaults and was corrected using actual producer settings, preserving the old run. The resource checker reconstructs both old LEFT intercepts on all 45 rows, then calculates 92 outward consequences with 12 rejection controls.

Independent reviews use separate worktrees and disclose construction roles. Their reports contain independent support algebra, curvature adversarial checks, alternative outward resource arithmetic, and the root's protocol review. Review agreement is not an additional mathematical premise.

The genuine base provider, equality-compatible attainment, same-law contact and moment analysis, strict confinement, Fourier bounds, remaining local acceptances, zero-noise Bernoulli input, lower approximation and complete composition contract remain obligations. No full LOCAL42, whole-proof PASS, or proof of the conjecture is asserted here.